BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 257 · Input and output devices

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Why is a GPU suitable for applying the same filter to many independent pixels?

Q2. What role does a processor's control unit have?

Q3. What data type should represent the state of a binary switch?

Q4. ASCII encodes A as decimal 65 and B as 66. A string 'ABBA' uses four 8-bit stored code units in this simplified example. How many bits and bytes, and why is this not proof all Unicode characters fit in one byte?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

sensor — Input transducer measuring a physical quantity.

ADC — Analogue-to-Digital Converter.

actuator — Output component that causes physical action.

latency — Delay between input and system response.

accessibility — Design usable by people with varied needs.

New learning

Select suitable input — A temperature sensor measures a physical quantity; an analogue-to-digital converter is needed if the controller uses digital samples.

Process and control — Software compares the sampled temperature with a target to decide whether to open the vent.

Choose appropriate outputs — An actuator makes the physical change; a display can show readings. Design must address accessibility and suitable latency.

Worked example

Choose input and output for an automatic door that opens when someone approaches.

1. Predict first — A proximity or motion sensor detects someone approaching and passes a signal to the controller.

2. Explain and check — A motor and driver create movement; safety sensing reduces the risk of closing onto a person.

We Do · Practise with guidance

Choose devices for an accessible pedestrian crossing with a push button, lights and audible alert.

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Button → controller → lights + audio/tactile signals
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Predict first

A push button is a clear intentional input and the controller processes it under safe traffic timing rules.

Explain and check

Outputs include signal lights plus appropriate audible or tactile cues to improve accessibility.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Name a suitable input and output device for a weather station, and explain the role of each.

Question 2

A greenhouse opens a vent when temperature exceeds a limit. Name the sensor, processing decision and actuator, and explain any analogue conversion.

Question 3

Evaluate device choices for a pedestrian crossing serving users with hearing or visual impairment.

Exit Ticket · Answer before checking

A motor that physically opens a door is an input device.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. It can apply similar computations to many data elements in parallel.

Q2. It decodes and coordinates instruction execution.

Q3. Boolean.

Q4. 4×8=32 bits=4 bytes. Unicode contains many more characters, and UTF encodings may use more than one byte for a character; 8-bit storage here is an assumption.

You Do pages · Independent questions

Q1. [4] Temperature sensor measures temperature [2]; display or data logger presents/stores readings [2].

Q2. [6] Temperature sensor [1] measures physical quantity [1]; ADC where needed [1]; controller compares with threshold [1]; motor actuator [1] opens vent [1].

Q3. [6] Button input should be accessible [1]; visual indications [1]; audible signal [1]; tactile rotating cone/vibration [1]; suitable signal timings [1]; prevent misleading or unsafe outputs [1].

Exit Ticket · Model answer

False. It is an output actuator.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first.

GREEN PEN Then: tick correct answers; rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding